

Homework #5:

Covers: Module 2.3-2.4 (Sections: 5.1-5.3)

D 5.44 Design the circuit of Fig. P5.44 to establish a drain current of 0.1 mA and a drain voltage of $+0.3\text{ V}$. The MOSFET has $V_t = 0.5\text{ V}$, $\mu_n C_{ox} = 400\text{ }\mu\text{A/V}^2$, $L = 0.4\text{ }\mu\text{m}$, and $W = 5\text{ }\mu\text{m}$. Specify the required values for R_S and R_D .

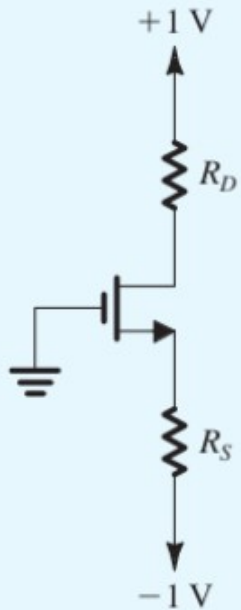


Figure P5.44

D 5.51 The NMOS transistors in the circuit of Fig. P5.51 have $V_t = 0.5\text{ V}$, $\mu_n C_{ox} = 90\text{ }\mu\text{A/V}^2$, $\lambda = 0$, and $L_1 = L_2 = L_3 = 0.5\text{ }\mu\text{m}$. Find the required values of gate width for each of Q_1 , Q_2 , and Q_3 to obtain the voltage and current values indicated.

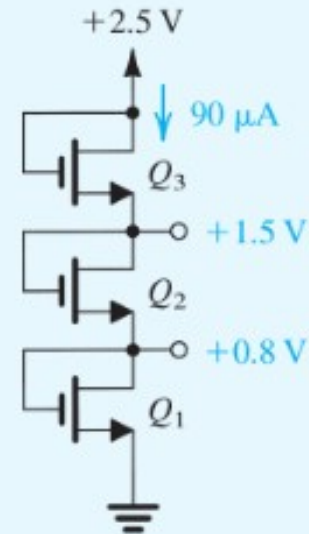


Figure P5.51

Show all steps, clearly state and verify all assumptions made. For PSpice simulation show the output file with the code and required output quantities. You may be asked to solve the assigned homework problems in class.

5.57 For each of the circuits shown in Fig. P5.57, find the labeled node voltages. The NMOS transistors have $V_t = 0.9 \text{ V}$ and $k'_n(W/L) = 1.5 \text{ mA/V}^2$.

Show PSpice simulation.

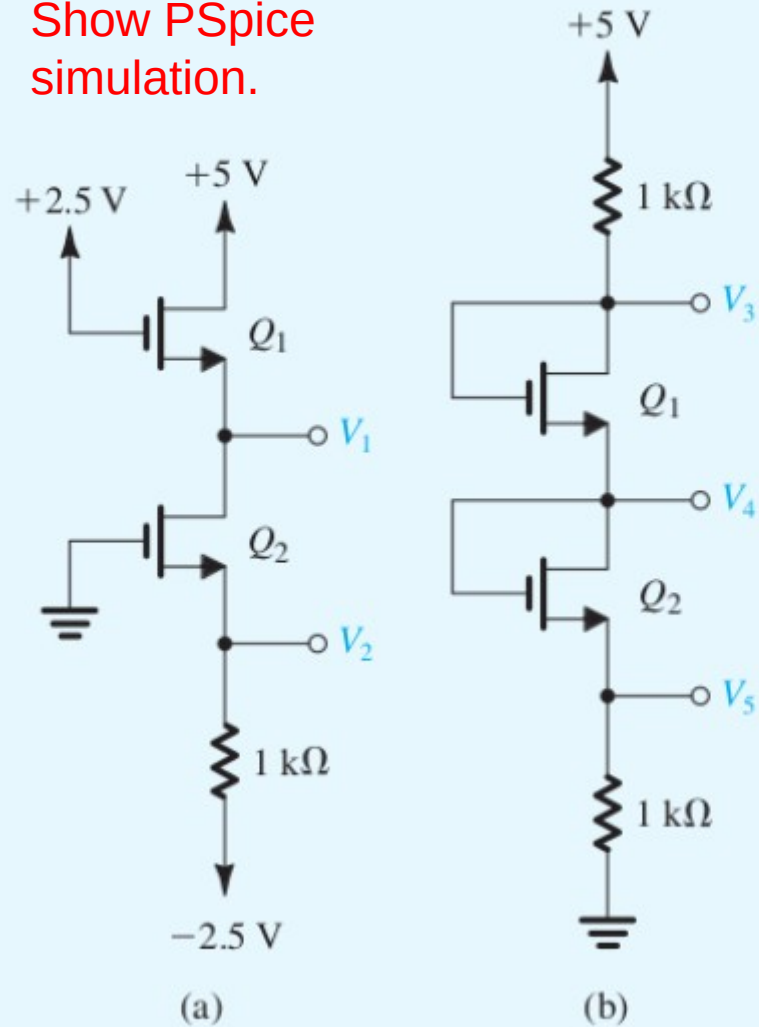


Figure P5.57

5.61 In the circuit of Fig. P5.61, transistors Q_1 and Q_2 have $V_t = 0.7 \text{ V}$, and the process transconductance parameter $k'_n = 125 \mu\text{A/V}^2$. Find V_1 , V_2 , and V_3 for each of the following cases:

- (a) $(W/L)_1 = (W/L)_2 = 20$
- (b) $(W/L)_1 = 1.5(W/L)_2 = 20$

Show PSpice simulation.

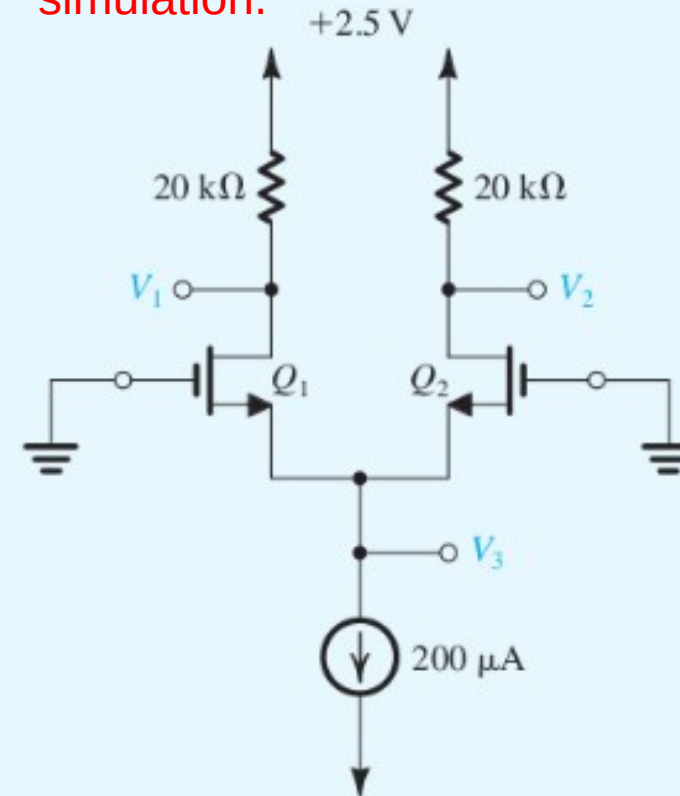


Figure P5.61